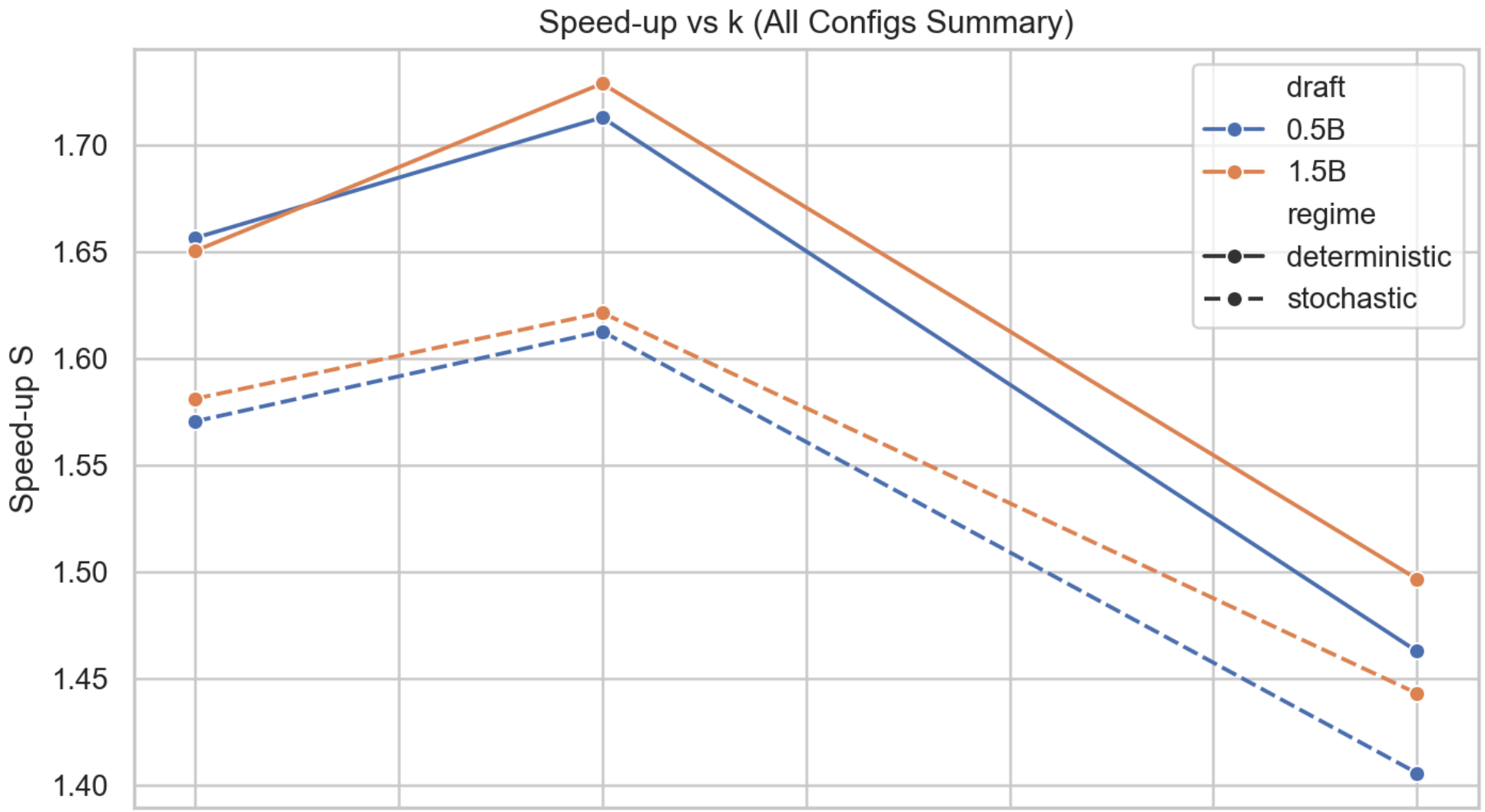
Speculative Decoding

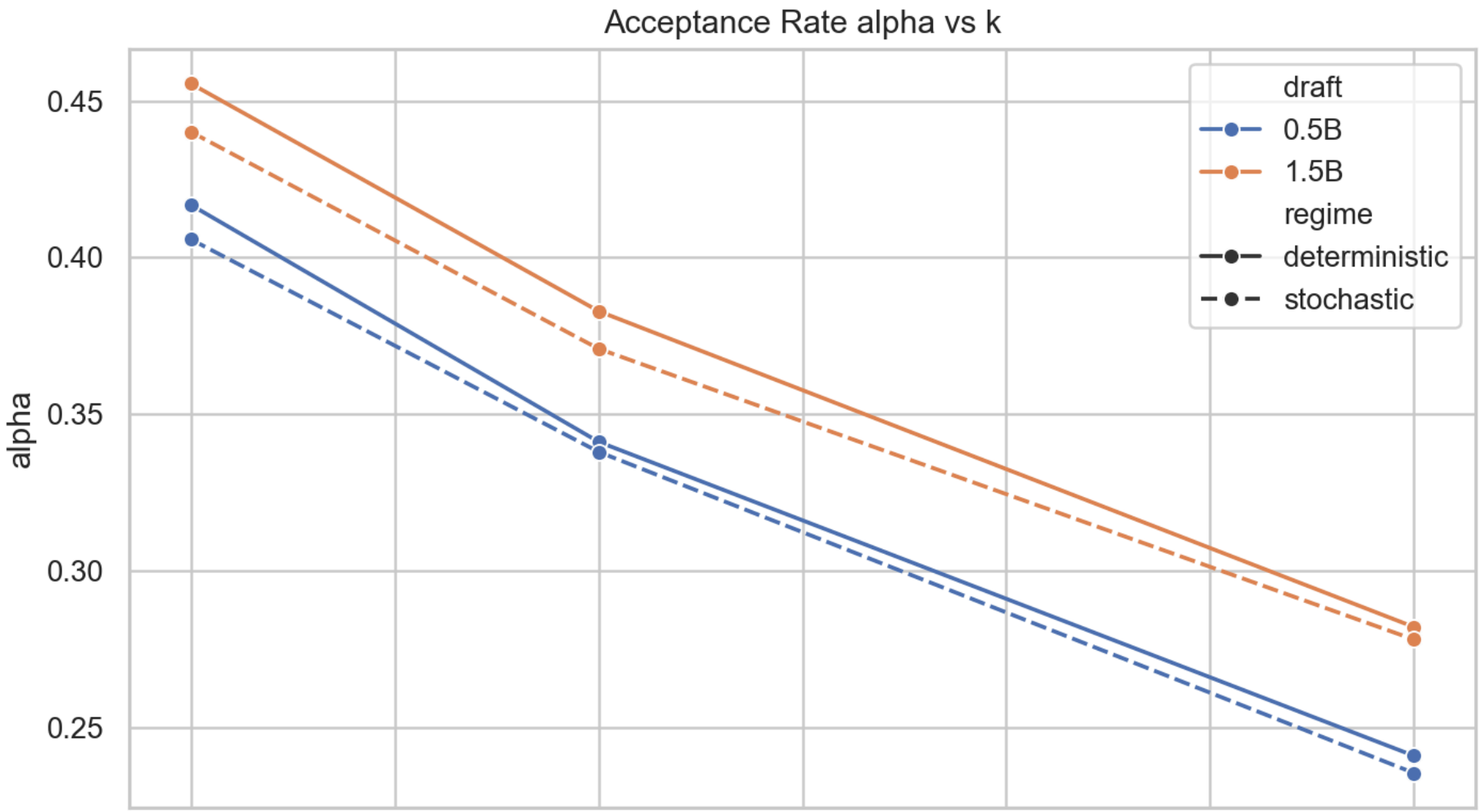
Visual diagnosis of wall time and speed-up limits

- Data source: `results/all_configs_summary.csv`, `results/pivot_time_breakdown_debug_quickrun.csv`, and compact ablation outputs
- Focus: `alpha`, `B_eff`, `k`, and stage-wise bottlenecks

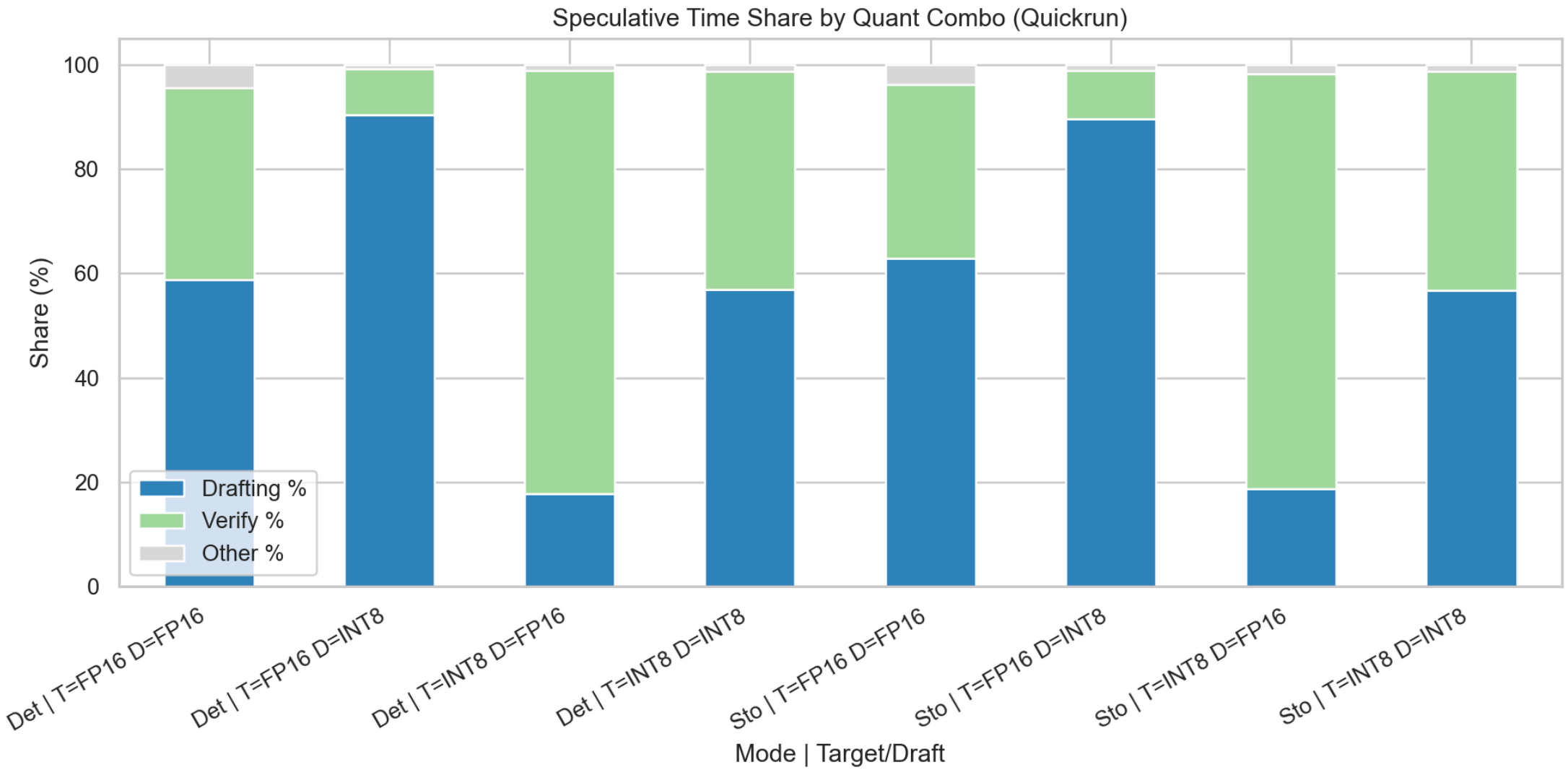
What We Observe First



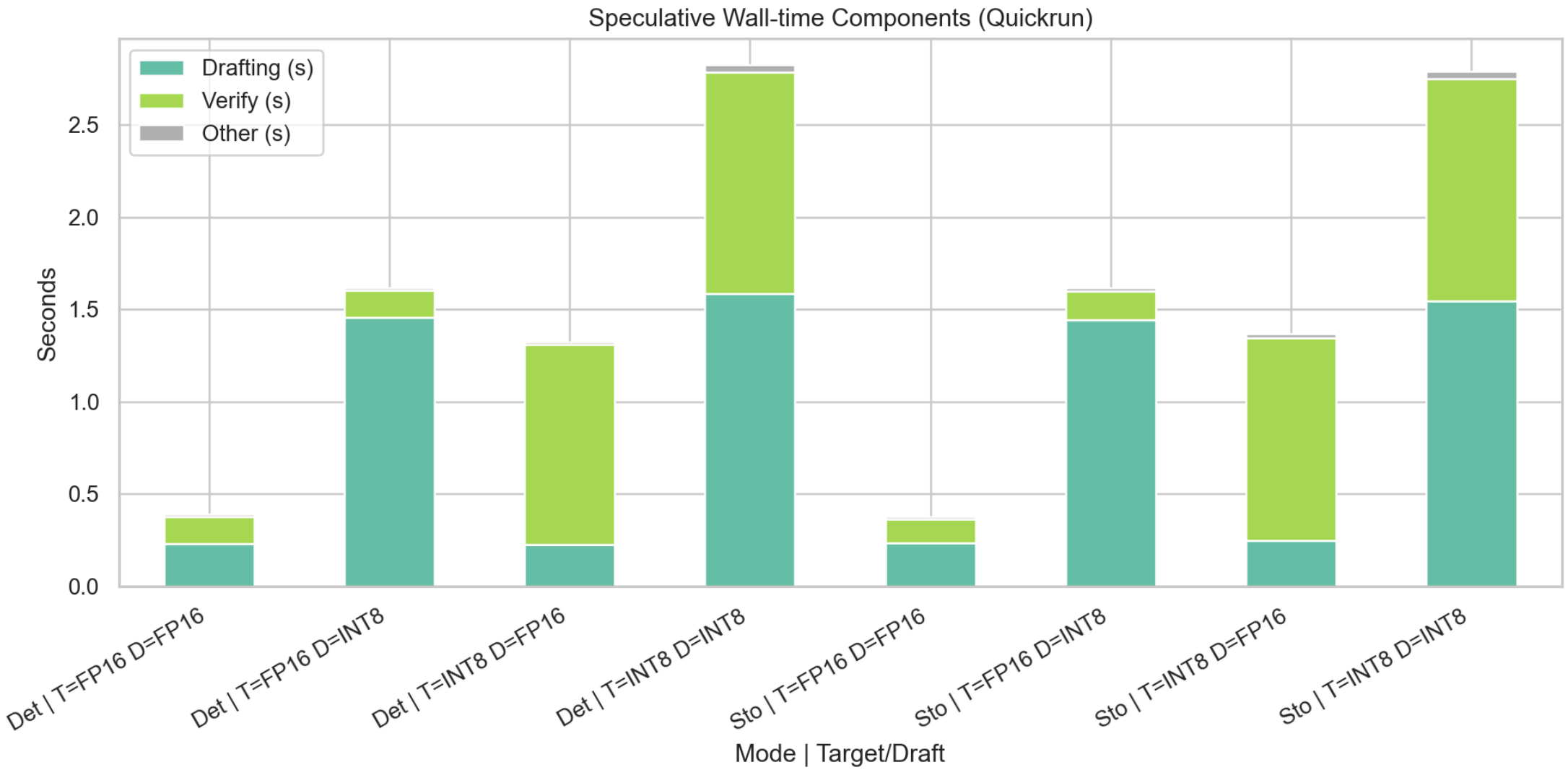
Acceptance Dynamics



Where Time Is Spent (Share)



Where Time Is Spent (Seconds)



Metric Lens

$$\alpha = \frac{\text{accepted}}{\text{proposed}}, \qquad B_{\text{eff}} = \frac{\text{accepted}}{\text{verify steps}}$$

$$S = \frac{T_{\text{baseline}}}{T_{\text{spec}}}, \qquad T_{\text{spec}} \approx T_{\text{draft}} + T_{\text{verify}} + T_{\text{other}}$$

- Even good `alpha` can underperform if draft/verify cost is imbalanced.

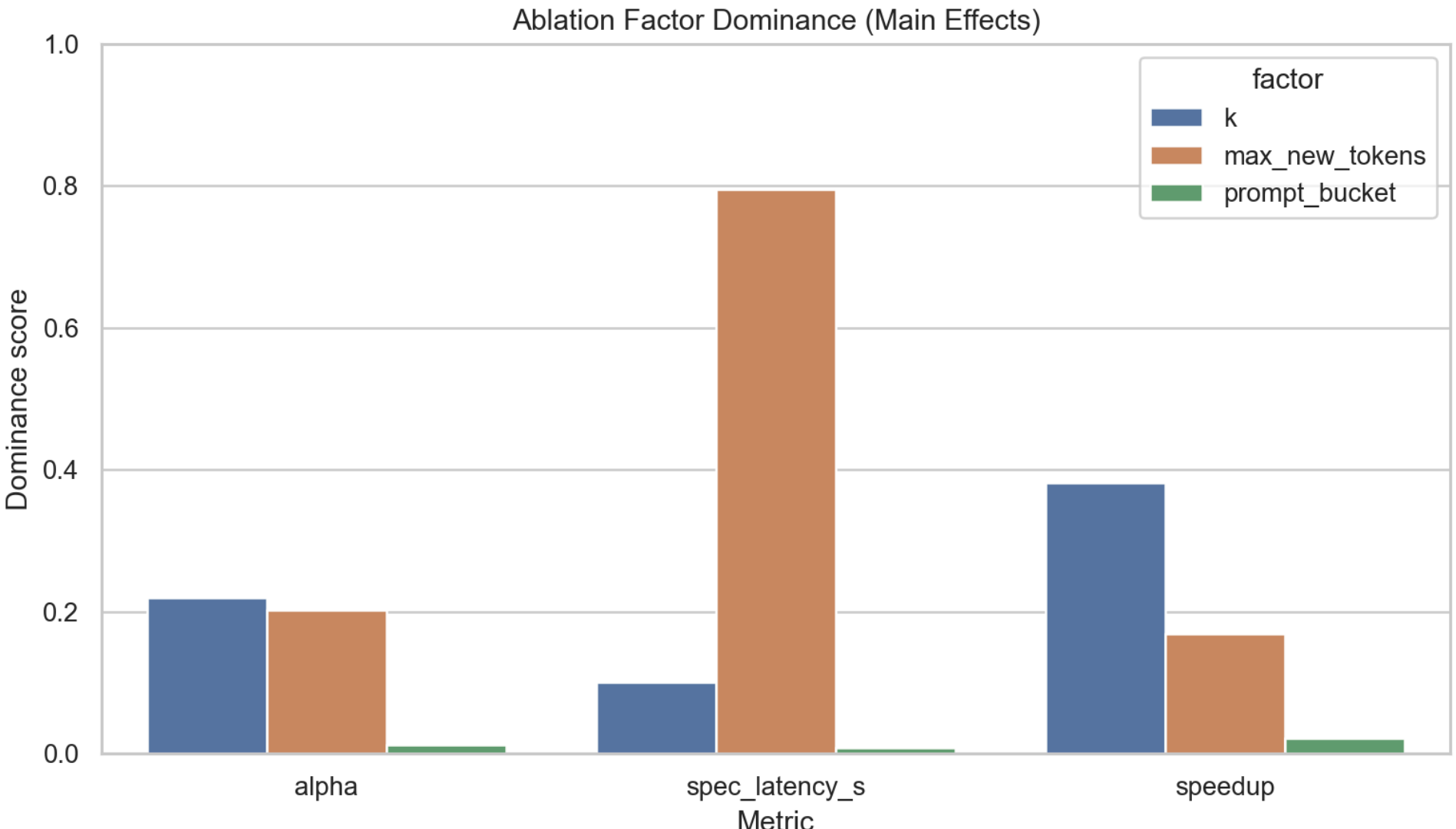
Why FP16 + FP16 Can Still Underperform

- Moderate acceptance still causes correction overhead.
- Draft path and verify path can remain serialized on one GPU.
- Fixed costs (prefill, synchronization, kernel launch) dominate short outputs.
- `k` above optimum increases wasted draft work.

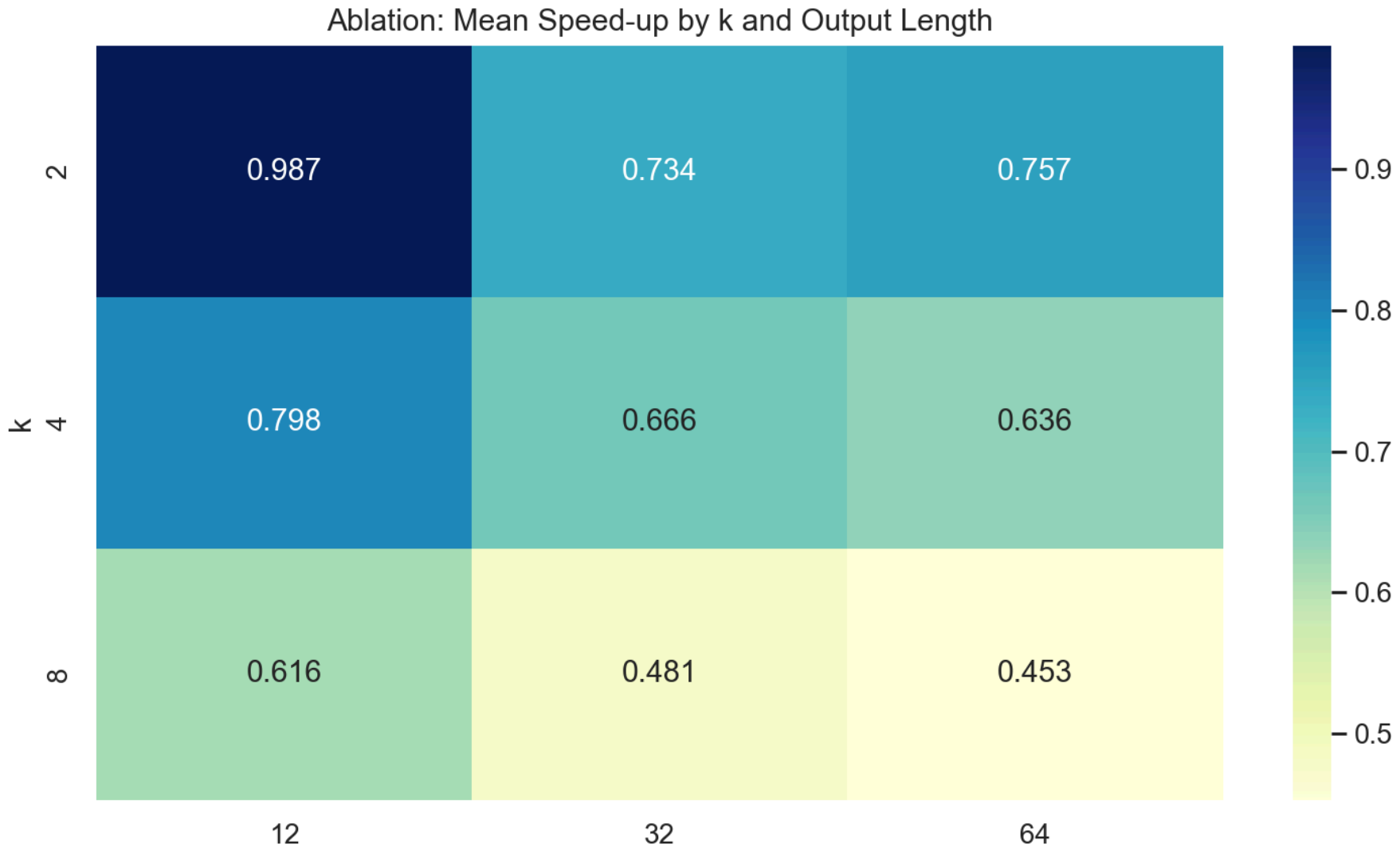
Practical Implication

- Optimize the system, not a single precision knob.
- Priorities:
  i. Find the local-optimal `k`.
  ii. Minimize dominant stage (draft or verify).
  iii. Benchmark on longer decode lengths to reduce fixed-cost distortion.
  iv. Track per-stage shares together with `alpha` and `B_eff`.

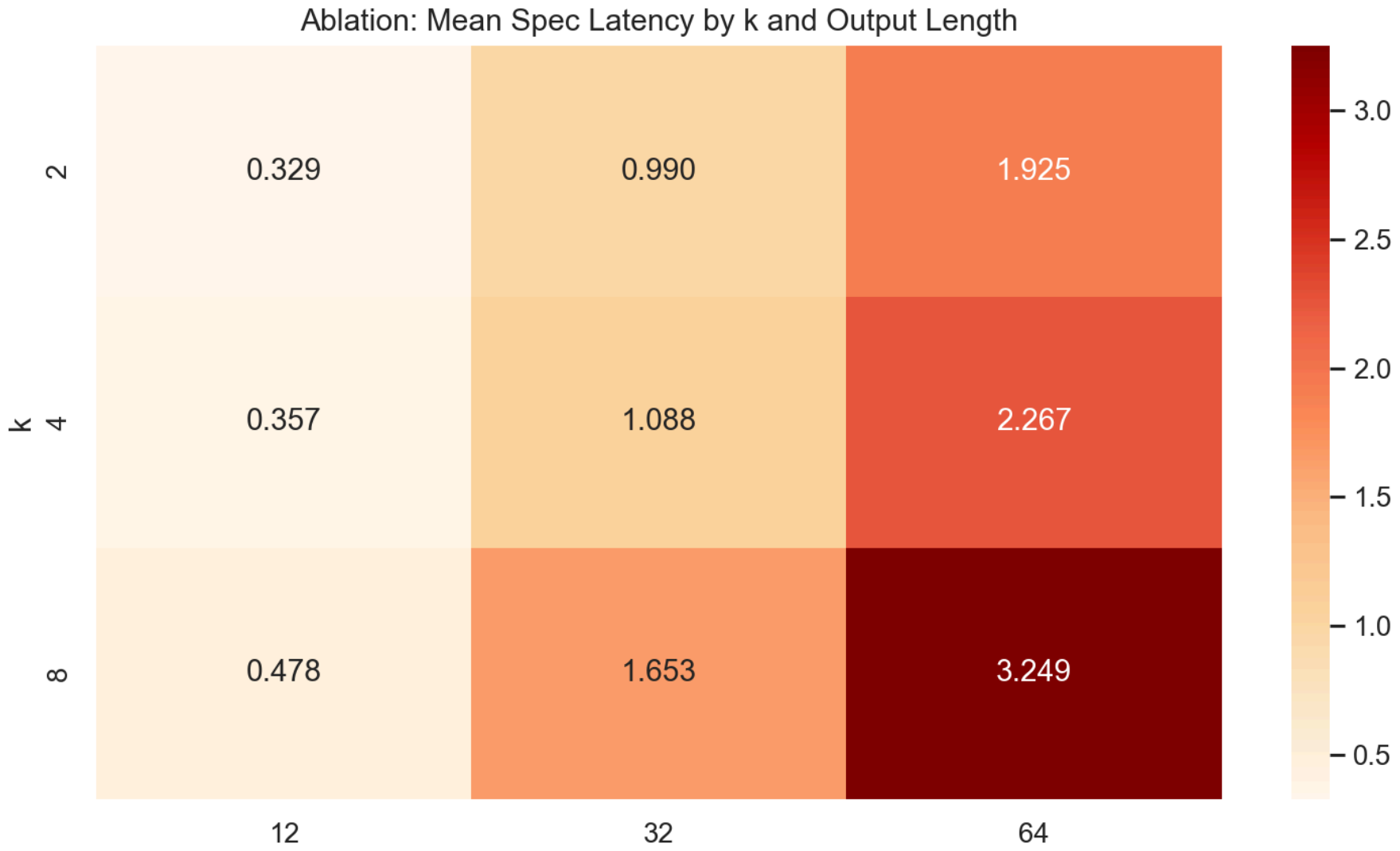
Compact Ablation: What Dominates?



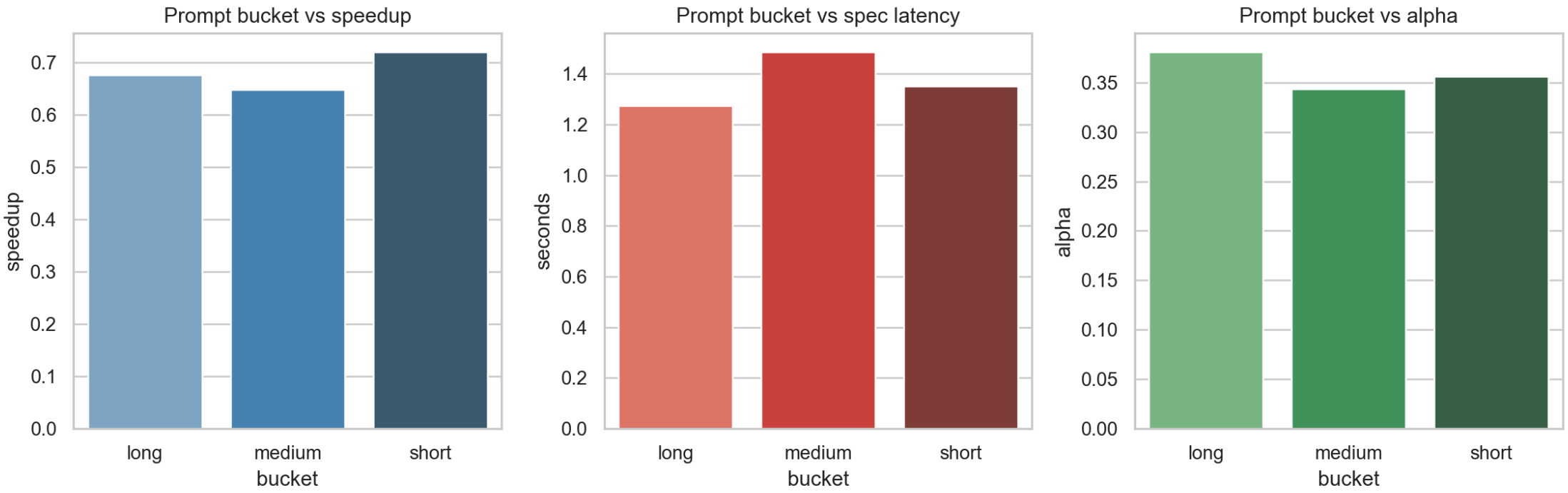
Ablation Heatmap: Speed-up



Ablation Heatmap: Spec Latency



Prompt Length Effect (Short vs Medium vs Long)



- Prompt bucket impact is secondary compared with `k` and output length.
- In this dataset, bucket changes are visible but not dominant.

Updated Takeaway (with Ablation)

- For FP16 target + FP16 draft, low speed-up is not mainly a precision issue.
- Dominant factors from this ablation:
  i. `k` controls speed-up behavior.
  ii. output length controls wall time.
  iii. prompt length has minor contribution (in this compact setup).
- Practical choice: keep `k` moderate and benchmark at representative output lengths.